Diabetes Multi-Class Classification & Clustering

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1. Introduction

This is a report on the end-to-end machine learning pipeline created to classify diabetes into multiple classes and unsupervised clustering on the Mendeley Diabetes Dataset. It was aimed at establishing models that differentiate three patient groups of Non-Diabetic (N), Pre-Diabetic (P), and Diabetic (Y) in the presence of 10 clinical biomarkers and patient demographics.

1.1 Dataset

Property	Details
Source	Mendeley: data.mendeley.com/datasets/wj9rwkp9c2/1
Total Samples	1,000 patients
Features	14 columns, 2 identifiers, 1 categorical (Gender), 10 clinical features, 1 target (CLASS)
Target Classes	N = Non-Diabetic (103), P = Pre-Diabetic (53), Y = Diabetic (844)
Missing Values	None, clean dataset

1.2 Clinical Features

Feature	Full Name	Clinical Significance
HbA1c	Glycated Hemoglobin	Primary diabetes diagnostic marker, reflects 3-month blood sugar average
BMI	Body Mass Index	Key obesity indicator, strong predictor across all classes
AGE	Patient Age	Older age associated with higher diabetes risk
Urea	Blood Urea Nitrogen	Kidney function marker; mildly elevated in diabetics
Cr	Creatinine	Kidney filtration marker, extreme outliers present (max=800)
Chol	Total Cholesterol	Lipid panel marker, weak separator between classes
TG	Triglycerides	Elevated in metabolic syndrome and diabetes
HDL	High-Density Lipoprotein	Protective cholesterol, slightly lower in diabetics
LDL	Low-Density Lipoprotein	Bad cholesterol, weak separator in this dataset
VLDL	Very Low-Density Lipoprotein	Strongly elevated in diabetics (0.94 - 2.02)
Gender	Categorical (F/M)	May influence metabolic parameters like BMI

2. Exploratory Data Analysis

Before building any models, the data was carefully examined to understand its structure, identify quality issues, and surface patterns that would guide modeling decisions.

2.1 Data Quality Findings

- No missing values: There are no absent values in all the 14 columns, which is unusual with clinical data.
- String inconsistencies: CLASS and Gender columns had trailing whitespace and inconsistent casing (e.g. 'Y ', 'f'). Modeling is done after being stripped and modeled.
- Duplicate patient visits: 39 patients were recorded with a number of visits. There was no actual duplication of rows, but the same patient is found in both the train and test sets. More accurate assessment would be provided by a split by patient ID in a group.
- There were unnecessary columns: ID and No_Pation were removed - no predictive value.

2.2 Class Distribution

The dataset is severely imbalanced, which is typical of clinical data on diabetes (most patients presenting are already diagnosed):

Class	Label	Count	Proportion
Y	Diabetic	844	84.4%
N	Non-Diabetic	103	10.3%
P	Pre-Diabetic	53	5.3%

This disproportion needs the use of SMOTE (Synthetic Minority Oversampling Technique).

2.3 Univariate Analysis: Key Findings

- Shapes of distribution: HbA1c and BMI are skewed to the right. Cr and VLDL possess very tails, which is proven by outlier analysis.
- Non-normality: The features are mostly not normally distributed, the tree based models can easily accommodate this.

2.4 Outliers

Identified with the help of the IQR method:

Feature	Outlier Count	Max Value	Handling
Cr	52	800	Clipped to 5th-95th percentile range
VLDL	74	35.0	Clipped, extreme outliers, but medically valid
Urea	65	38.9	Clipped to realistic clinical range

The outliers have not been removed but instead clipped since in clinical data, the outliers may be actual. A 800 Creatinine is out of the ordinary but medically valid (severe kidney disease). Elimination of such patients will create bias.

2.5 Mean Feature Values by Class

Feature	N (Non-Diab)	P (Pre-Diab)	Y (Diabetic)	Insight
HbA1c	4.56	6.00	8.88	Clearest separator, follows WHO thresholds exactly
BMI	22.37	23.93	30.81	100% of obese patients are diabetic in this dataset
VLDL	0.94	0.98	2.02	Doubles in diabetics, strong metabolic signal
TG	1.63	2.13	2.45	Progressive rise with diabetes severity
AGE	44.23	43.28	55.31	Diabetics are 11 years older on average
HDL	1.23	1.13	1.21	Weak separator, minimal class difference
LDL	2.63	2.49	2.62	Near-identical, poor discriminator
Chol	4.27	4.58	4.95	Mild upward trend, weak signal

2.6 HbA1c - Primary Diagnostic Characteristics.

We perform dedicated analysis on HbA1c because it is the primary WHO-recognized diabetes marker.

Class	Count	Mean	Std	Min	25%	Median	Max
N	103	4.56	0.92	0.9	4.1	4.9	5.6
P	53	6.00	0.19	5.7	5.9	6.0	6.4
Y	844	8.88	2.26	2.0	7.2	8.8	16.0

- Pre-Diabetic: P class std = 0.19 (very tight) - tightly concentrated about the clinical edge of 6.0.
- Diabetic: Y class std = 2.26 (high) - there is a wide range of severity in diabetic patients.

2.7 BMI Analysis

BMI was almost an ideal separator of extreme classes:

BMI Category	N (%)	P (%)	Y (%)
Normal	47	22	31
Overweight	0	1	99
Obese	0	0	100

Obese patients are all diabetic, 99 percent of overweight patients are diabetic. This almost absolute BMI division is not typical and partially might be due to the composition of the datasets.

2.8 Gender Analysis

Gender is statistically significantly correlated with diabetes class. The cases of diabetic males are higher (490 vs 354 females although there were fewer males). The feature of gender was not eliminated.

2.9 LDL/HDL Ratio

The LDL/ HD ratio indicated almost the same values in every class: N=2.49, P=2.40, Y=2.56. This ratio contributes a degree of little discrimination to classification in this dataset.

2.10 Correlation Findings

- Highest correlation: Urea and Cr: r = 0.62 (hypothesized - both kidney markers).
- Moderate correlations: BMI and HbA1c: r = 0.41. HbA1C and AGE: r = 0.38, both clinically significant.
- General: The majority of feature pairs are weakly correlated with each other. None of the features are severely multicollinear and demand elimination.

3. Classification Pipeline

Classifications were trained and assessed using four models, and the hyperparameter tuning using Optuna was applied to the most successful one.

3.1 Preprocessing Pipeline

- Data cleaning: Trimming of strings and standardizing of cases in CLASS (5 unique to 3) and Gender (3 unique to 2).
- Column dropping: NoPation and ID columns dropped (not predictive).
- Outlier treatment: Percentile clipping (5 th -95 th) of Urea, Cr, and VLDL to minimize the influence of outliers without eliminating actual clinical data.
- Train/test split Stratified 80/20 split: Train = 800 samples (Y:675, N:82, P:43), Test = 200 samples (Y:169, N:21, P:10).
- Prevention of leakage: Split is achieved prior to any scaling or encoding to ensure that data leakage is prevented.

3.2 Feature Encoding

Feature Type	Encoder	Details
Target (CLASS)	LabelEncoder	N=0, P=1, Y=2
Gender	OrdinalEncoder	F=0, M=1
10 Numerical Features	StandardScaler	Zero mean, unit variance, fit on train only

Every change is enclosed in a ColumnTransformer within an ImbPipeline (imbalanced-learn) such that SMOTE does not operate on the whole dataset. This is critical as fitting SMOTE before splitting would create synthetic test samples from training data (data leakage).

3.3 Model Results

Model	Accuracy	Macro F1	Weighted F1	Errors
Random Forest	0.995	0.9913	0.9951	1
XGBoost (baseline)	0.995	0.9909	0.9949	1
XGBoost (tuned)	0.995	0.9909	0.9949	1
Decision Tree	0.980	0.9128	0.9776	4
Logistic Regression	0.955	0.8999	0.9574	9

3.4 Per-Class F1 Scores

Model	N F1	P F1	Y F1	Macro Avg
Logistic Regression	0.89	0.83	0.97	0.90
Decision Tree	1.00	0.89	0.99	0.96
Decision Tree (Tuned)	1.00	0.75	0.99	0.91
Random Forest	0.98	1.00	1.00	0.99
XGBoost	0.98	1.00	1.00	0.99
XGBoost (Optuna)	0.98	1.00	1.00	0.99

The P (Pre-Diabetic) segment is the segment which is always the most difficult to segment in all the models F1 = 0.75-1.00. This should be seen clinically as by definition also pre-diabetic are both normal and diabetic.

3.5 Error Analysis

Best model (XGBoost Optuna): 1 total error on 200 test samples (0.5% error rate).

True Class	Predicted	HbA1c	BMI	AGE
N (Non-Diabetic)	Y (Diabetic)	4.2	24.0	59

- Analysis: HbA1c = 4.2 is in the normal range (mean correct = 8.22). A borderline value did not result in the model error.
- Possible reason: Model might behaved this due to interaction of other features (age, other biomarkers)

- Critical finding: No critical error. There were no critical errors in which zero diabetic patients (Y) were classified as non-diabetic (N). Zero errors were in the clinically dangerous direction.

3.6 Hyperparameter Tuning: Optuna

Optuna (Bayesian optimization) was used instead of grid search. It learns which parameter regions are promising and focuses trials there, making it far more efficient than exhaustive search.

Hyperparameter	Optimal Value	Notes
classifier	XGBClassifier	Outperformed RF, GBM, LR across trials
n_estimators	277	Moderate depth ensemble
learning_rate	0.1684	Conservative, reduces overfitting
max_depth	4	Shallow trees, good bias/variance balance
min_child_weight	5	Prevents splits on too few samples
subsample	0.9136	91% of rows per tree, near-full data use
colsample_bytree	0.9303	93% of features per tree
reg_alpha	0.9176	L1 regularization, feature sparsity
reg_lambda	4.8058	L2 regularization, weight smoothing

Best CV Macro F1: 0.9686 (Trial Score: 0.968580880089231). XGBClassifier was chosen as the most successful algorithm in 80 Bayesian trials as selected by Optuna.

3.7 Model Strengths and Weaknesses

Model	Strengths	Weaknesses
Logistic Regression	Interpretable, fast, good baseline	Assumes linear boundaries, lowest performance
Decision Tree	Transparent rules, fast inference	Prone to overfitting without depth control
Random Forest	Robust, strong feature importance	Slower to train, less interpretable
XGBoost	Highest accuracy, handles class imbalance well	Sensitive to hyperparameters

4. Clustering Analysis

Although the labels exist, clustering was conducted to inquire whether unsupervised algorithm grouping would behave the same way clinicians group these patients.

4.1 Preprocessing for Clustering

- Feature scaling: Before clustering, all the 11 features were scaled using StandardScaler. Algorithms such as K-Means that are distance based demand equal hits.
- PCA variance: 2 PCs explain 36.4%, 3 PCs explain 51.7%, 5 PCs explain 72.9%. The reason why 2D cluster visualizations are lossy is due to high dimensionality.

4.2 Algorithm Results

Algorithm	Silhouette	Davies-Bouldin	ARI	NMI
K-Means (k=3)	0.1939	1.6545	0.4801	0.3928
Agglomerative (Ward)	0.1868	1.6643	0.4419	0.3203
DBSCAN	0.0163	1.0578	-0.0677	0.0832

Metric guide:
Silhouette Score (high = good, maximum=1)
Davies-Bouldin Index (low = good, minimum=0)
ARI (congruence with true labels, maximum=1)
NMI (mutual information with true labels, maximum=1).

4.3 K-Means Analysis

- Optimal k: Best K = 3, silhouette (0.194).
- Low silhouette clarified: The silhouette of 0.194 is low and anticipated. By definition, pre-diabetic patients lie between N and Y, and therefore, there are naturally soft cluster boundaries.
- ARI interpretation: ARI = 0.4801 implies that the algorithm labels items 48 percent more accurately than random, without even being exposed to the true labels. This is the assurance that the data is really separably structured.

4.4 K-Means Cluster Profiles

Cluster	HbA1c	BMI	VLDL	TG	AGE	Cr
Cluster 0	5.56	23.47	0.91	1.92	43.59	60.76
Cluster 1	8.67	32.53	7.98	2.21	48.03	69.93
Cluster 2	9.07	31.23	1.11	2.42	57.04	62.65

4.5 Why DBSCAN Failed

DBSCAN identified 9 clusters and the remaining 597 patients (59.7) were considered noise. This is not a parameter tuning failure, it reflects the data issue. Diabetes biomarkers form continuous gradients, not isolated dense islands. DBSCAN is the wrong algorithm for this problem.

4.6 Classification vs Clustering Comparison

Approach	Best Method	Accuracy	ARI
Supervised (Classification)	XGBoost (Tuned)	99.5%	-
Unsupervised (Clustering)	K-Means (k=3)	-	0.4801

Supervised classification dramatically outperforms unsupervised clustering, as expected. The value of clustering is independent validation (confirming data has separable structure) and subgroup discovery, not prediction performance.

5. Conclusion

The project was able to prove an end-to-end machine learning pipeline to classify diabetes and unsupervised patient profiling. Using XGBoost optimized using Optuna, the model performed close to perfect accuracy of 99.5 percent and Macro F1-score of 0.99, and more importantly, zero false normals (no diabetic patients were overlooked).

5.1 Key Findings & Validation

This report established that HbA1c is still the most effective diagnostic tool with the result of 91.7% of the thresholds set by the WHO. But the great predictive ability of BMI, with 100 percent of patients in this cohort obese leading to diabetic, implies the presence of a strong but possibly simplistic relationship in this particular dataset.

Independent validation on the structure of the data was made by the unsupervised clustering using K-Means (k=3). A clinically significant Adjusted Rand Index of 0.48 indicates that the medical biomarkers cluster patients into categories that are related to their medical labels even with the absence of the model showing the column of the Target.

5.4 Model Assessment

Extremely high scores (99%+) are probably inflated by patient data overlapping in the train and test sets. The dataset contains 39 patients with multiple visits. In production, a group-based split by patient ID would give a more honest evaluation of generalization to genuinely new patients.

5.5 Future Enhancements

- Data split: Group split on patient ID to avoid almost duplicate leakage.
- Additional data on P class- Current P class (53 samples) is severely underrepresented.
- Semi-supervised: Attempt semi-supervised learning, use structure of discovered clustering to direct classification.
- Explainability SHAP values per-prediction explainability. Useful in clinical deployment.
- External validation: Evaluate on external dataset to test how well it generalizes beyond this particular cohort.